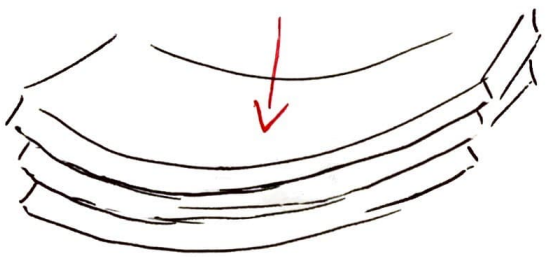
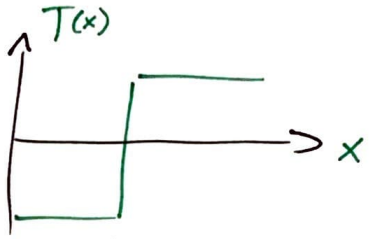
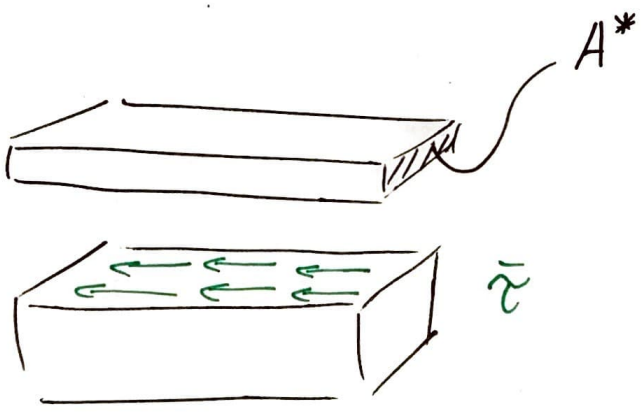
Övning 11 - Böjskjuvspänningar







Jämvikt måste uppfyllas

A^* - Area ovanför tvärsnittet

$$\bar{\tau} = \frac{T S_{A^*}}{I b} = \frac{1}{b}\frac{dF}{dx} \quad [FS\ 6.12] \quad \frac{dF}{dx} = \frac{T S_{A^*}}{I}$$

T - tvärkraft

S_{A^*} - Statiskt moment

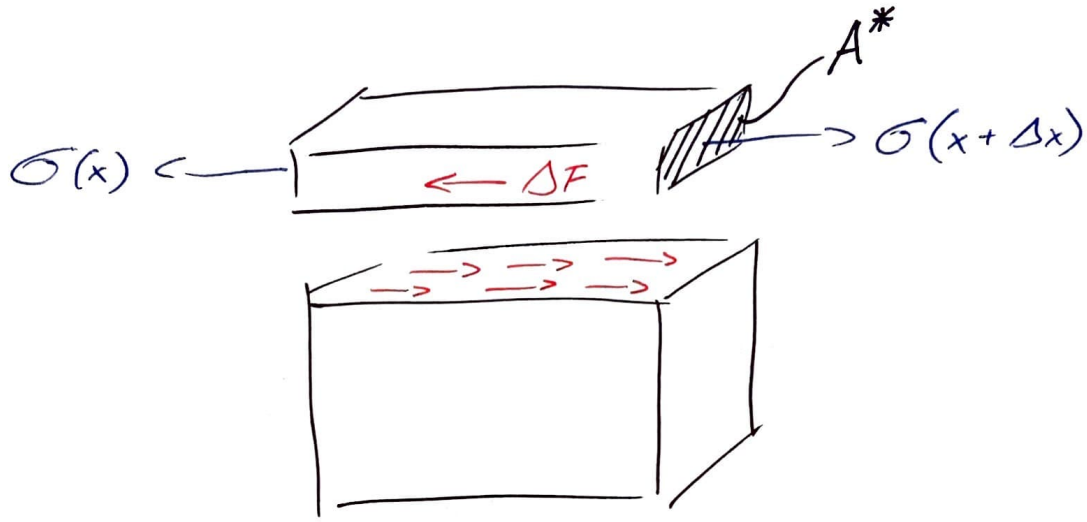
b - Bredd på snittyta

F - Skjuvkraft

$$S_{A^*} = \int_{A^*} z\,dA \quad [FS\ 30.1] \qquad S_\eta = z_{TP} \cdot A \quad [FS\ 30.4]$$

$$I_y = \int z^2\,dA \quad [FS\ 30.3]$$

Böjskuvspänning härledning



Jämvikt!

$$\rightarrow: \quad \int_{A^*} \sigma(x+\Delta x) - \sigma(x)\, dA^* - \Delta F = 0$$

$$\Delta F = \int_{A^*} \sigma(x+\Delta x) - \sigma(x)\, dA^*$$

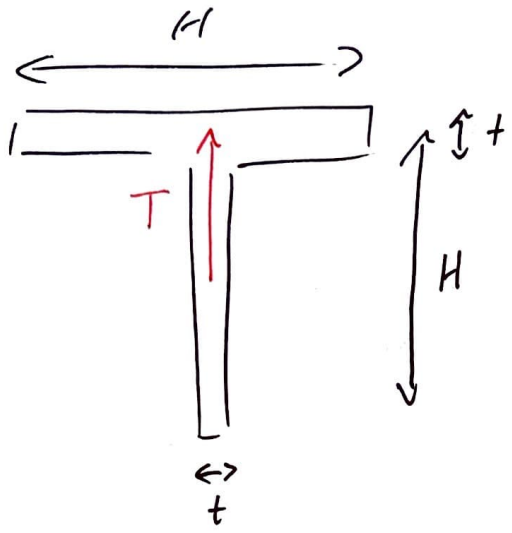
$$\left\{ \sigma = \frac{M}{I_y} z \right\} \Rightarrow \Delta F = \int_{A^*} \frac{\Delta M}{I_y} z\, dA^* = \frac{\Delta M}{I_y} \int_{A^*} z\, dA^*$$

$$\Rightarrow \Delta F = \frac{\Delta M\, S_{A^*}}{I_y}$$

$$\bar{\tau}\, b \Delta x = \frac{\Delta M\, S_{A^*}}{I_y} \Rightarrow \bar{\tau} = \frac{\Delta M}{\Delta x} \frac{S_{A^*}}{I_y b}$$

$$\Rightarrow \boxed{\bar{\tau} = \frac{T\, S_{A^*}}{I_y b}}$$

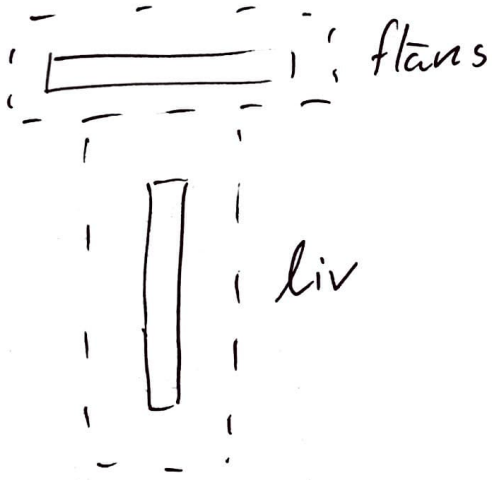
2.4:82



Tunnväggigt $t << H$

Tvärkraft T

Sökt: Skjuvspänningsdiagram över olika delar.

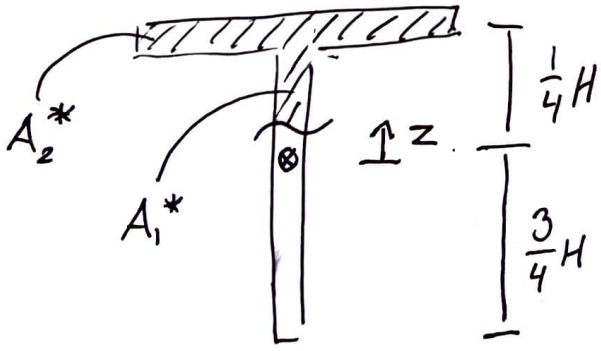


$$\bar{\tau} = \frac{T S_{A^*}}{I b}$$

Bestäm I_y från [Tab 30.2.10]

$$I_y = \frac{5 H^3 t}{24} \qquad e_z = \frac{3}{4} H$$

Bestäm S_{A^*} - Olika för fläns och liv

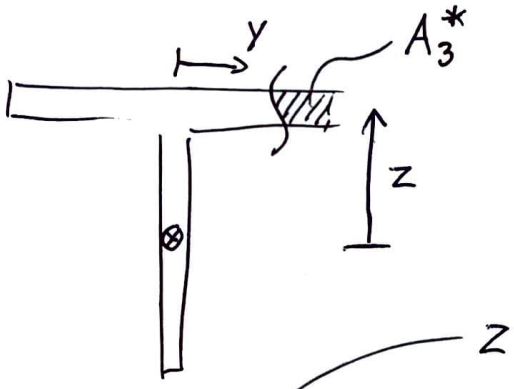


$$S_{A^*} = \sum_i S_{A_i^*} = \sum_i z_{TP} A_i$$

[FS 30.4]

2.4:82 forts

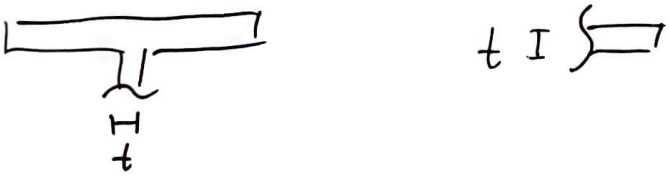
$$S_{A^*\ell} = \underbrace{\frac{H}{4}(Ht)}_{A_1^*} + \underbrace{\frac{\left(\frac{H}{4}+z\right)}{2}\left(\left(\frac{H}{4}-z\right)t\right)}_{A_2^*} = \frac{t}{2}\left(\frac{9H^2}{16}-z^2\right)$$



$$S_{A^*f} = \frac{H}{4}\left(\frac{H}{2}-y\right)t = \frac{Ht}{4}\left(\frac{H}{2}-y\right)$$

Bestäm b

$b_\ell = t$ $\qquad$ $b_f = t$



Beräkna böjskjuvspänning

$$\bar{\tau}_{liv} = \frac{T\,S_{A^*\ell}}{I_y\,b_\ell} = \frac{12T\left(\frac{9H^2}{16}-z^2\right)}{5Ht}$$

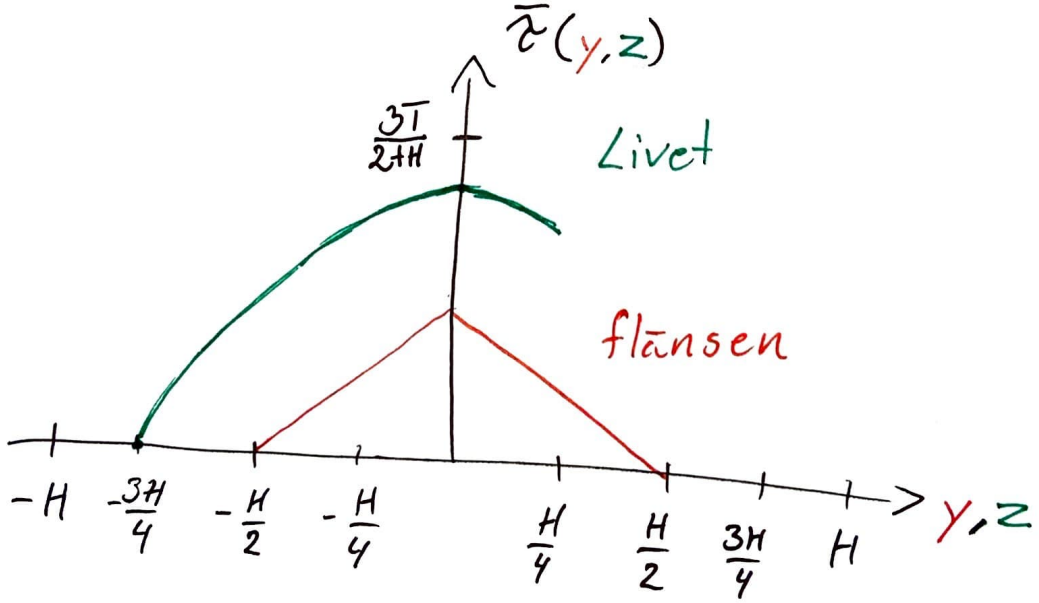
$$\bar{\tau}_{fläns} = \frac{T\,S_{A^*f}}{I_y\,b_f} = \frac{6T\left(\frac{H}{2}-y\right)}{5tH^2}$$

2.4:82 del 3

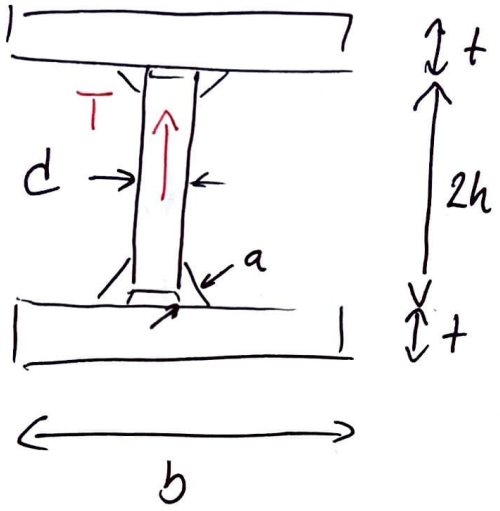
⑤ Spänningsdiagram

$$-\frac{3}{4}H < z < \frac{1}{4}H \Rightarrow \begin{cases} \bar{\tau}(-\frac{3}{4}H) = 0 \\ \bar{\tau}(0) = \frac{27T}{20tH} \\ \bar{\tau}(\frac{1}{4}H) = \frac{3T}{10tH} \end{cases}$$

$$-\frac{1}{2}H < z < \frac{1}{2}H \Rightarrow \begin{cases} \bar{\tau}(-\frac{1}{2}H) = 0 \\ \bar{\tau}(0) = \frac{3T}{5tH} \\ \bar{\tau}(\frac{1}{2}) = 0 \end{cases}$$



2.4:86



$I_y = 7 \cdot 10^7 \, mm^4$

$h = 100 mm$

$t = 20 mm$

$b = 140 mm$

$d = 10 mm$

$T = 10^5 N$

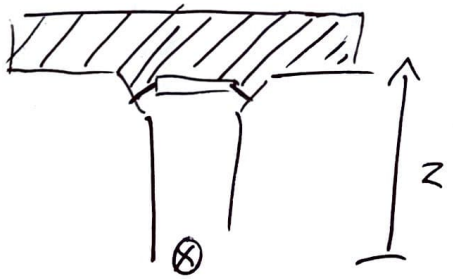
$a = 11 mm$

Sökt: Maximal böjskjuvspänning i svetsen

$$\bar{\tau} = \frac{T S_{A^*}}{I_y \bar{b}}$$

S_{A^*} och $\bar{b}$ okända

Bestäm statiska momentet S_{A^*}



$$A^* = A^*_{fläns} + A^*_{svets}$$

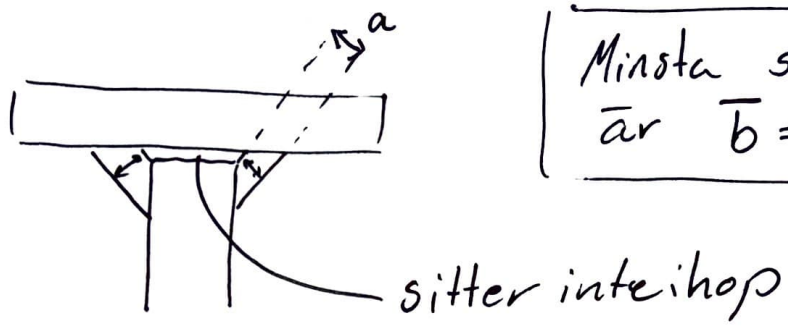
$$A^* \approx A^*_{fläns}$$

$$S_{A^*} = (2h+t)\frac{1}{2}bt = \frac{bt}{2}(2h+t)$$

$$S_{A^*} \approx 3.08 \cdot 10^5 \, mm^3$$

2.4:86 del 2

② Bestäm $\bar{b}$



Minsta snittet i svetsen är $\bar{b} = 2a$

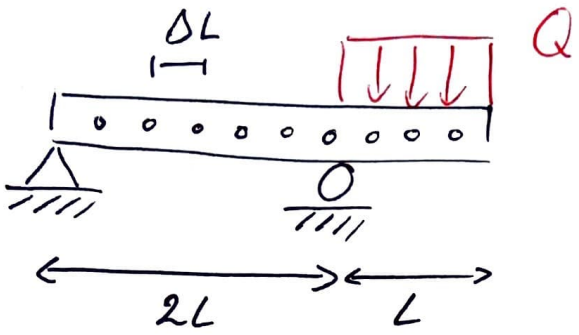
③ Beräkna böjskjuvningen

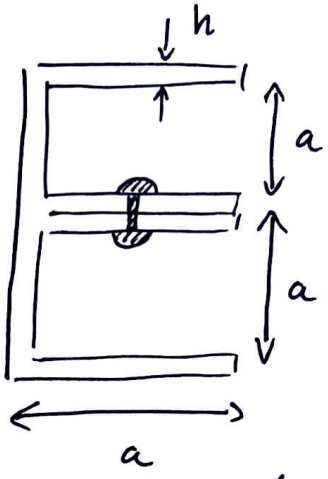
$$\bar{\tau} = \frac{T S_{A^*}}{I_y b} = \frac{T \cdot 3.08 \cdot 10^5 \, mm^3}{I_y \; 2a}$$

$$= \frac{10^3 \, N \cdot 3.08 \cdot 10^5 \, mm^3}{7 \cdot 10^7 \, mm^4 \cdot 2 \cdot 11 \, mm} = 20 \, N/mm^2$$

$$\bar{\tau} = 20 \, MPa$$

2.4:88



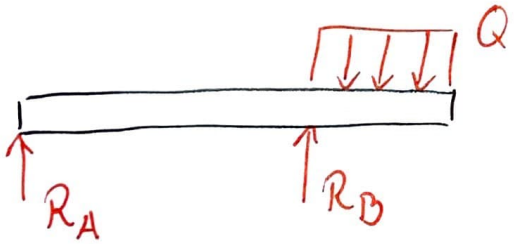


Tunnväggigt

Sökt: Hur stor last Q krävs för att skjuvkraft F_{nit} uppnås i en nit.

Böjskjuvkraften kräver tvärkraften i balken

① Bestäm $T(x)$ via frilägging



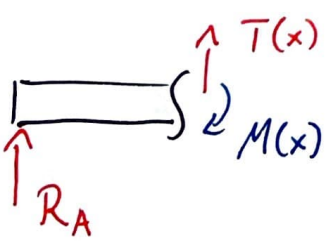
$$\uparrow: R_A + R_B - Q = 0$$

$$A: Q\frac{5L}{2} - R_B 2L = 0$$

$$\Rightarrow \begin{cases} R_A = -Q\frac{L}{4} \\ R_B = Q\frac{5L}{4} \end{cases}$$

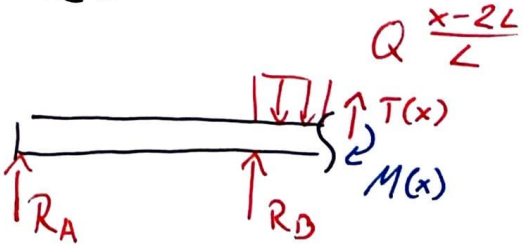
Snittning

$0 < x < 2L$



$$T(x) = -R_A = \boxed{Q\frac{L}{4}}$$

$2L < x < 3L$



$$T(x) = -R_A - R_B + Q\frac{x-2L}{L} = \boxed{Q\frac{x-3L}{L}}$$

2.4:88 del 2

$$\frac{dF}{dx} = \frac{T S_{A^*}}{I_y}$$

S_{A^*} och I_y okända

Bestäm I_y

$$I_y = 2\sum_i \left(I_{y_i} + z_i^2 A_i\right)$$

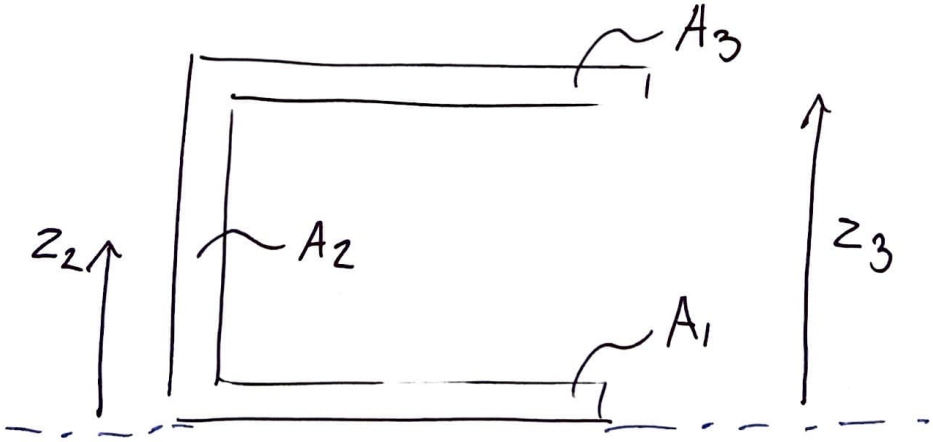
$$= 2\left(\frac{ah^3}{12} + \frac{ha^3}{12} + \frac{a^2}{4}ah + \frac{ah^3}{12} + a^2 ah\right)$$

$$= 2\left(0 + \frac{ha^3}{12} + \frac{ha^3}{4} + 0 + a^3 h\right)$$

$$= \frac{8a^3 h}{3}$$

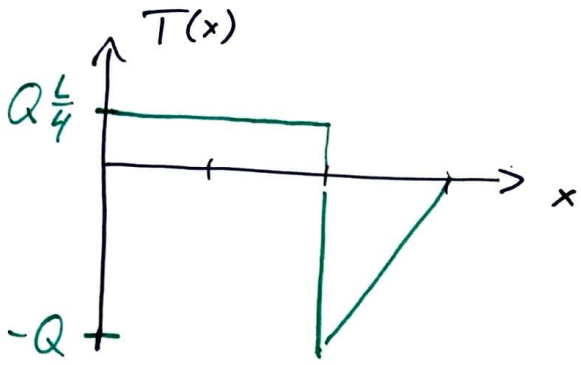
Bestäm S_{A^*}

$$S_{A^*} = \sum z_i A_i = \frac{a}{2}ah + aah = \frac{3a^2 h}{2}$$



2.4:88 del 3

③ Rita upp $T(x)$ och bestäm max



Max vid $2L$

④ Bestäm F för $\boxed{2L < x < 2L+\Delta L}$

$$\frac{dF}{dx} = \frac{T S_{A^*}}{I_y} \quad [FS\ 6.11]$$

$$\frac{dF}{dx} = \frac{Q\frac{x-3L}{L}\ \frac{3}{2}a^2h}{\frac{8}{3}a^3h} = \frac{9}{16aL}Q(x-3L)$$

$$F = \int_{2L}^{2L+\Delta L} \frac{9}{16aL}Q(x-3L)\,dx = \frac{9Q}{16aL}\left[\frac{x^2}{2} - 3Lx\right]_{2L}^{2L+\Delta L}$$

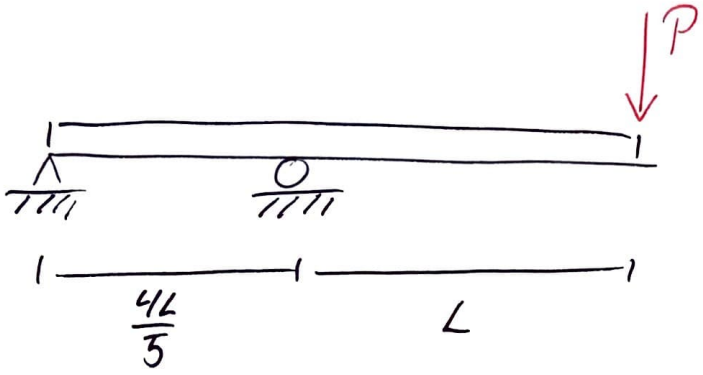
$$F = \frac{9Q}{16aL}\left(\frac{\Delta L^2}{2} - L\Delta L\right) \qquad \Delta L^2 \approx 0$$

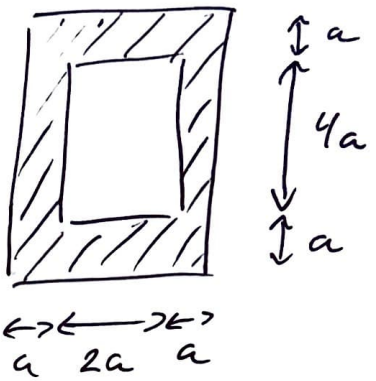
$$F = -\frac{9Q}{16a}\Delta L \qquad |F| = F_{nit} \text{ när } Q = Q_{max}$$

$$F_{nit} = \frac{9Q_{max}}{16a}\Delta L \Rightarrow \boxed{Q_{max} = \frac{16\,F_{nit}\,a}{9\,\Delta L}}$$

Samma svar om $T = Q$

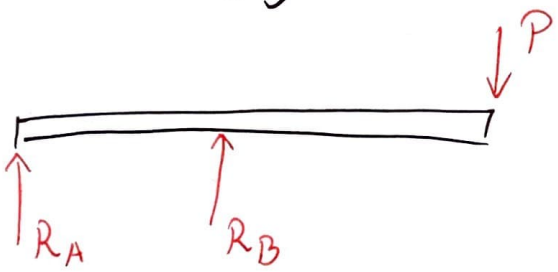
2015-01-17 Uppgift 4





Sökt: Största böjskjuvspänningen, och var.

① Frilägg



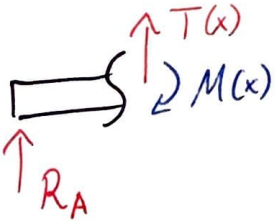
② Jämvikt

$\uparrow: R_A + R_B - P = 0$

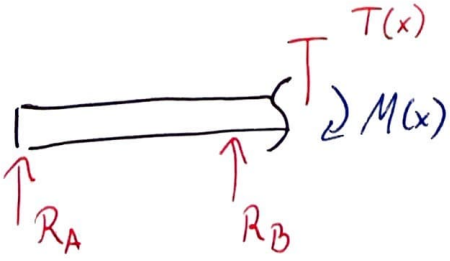
$A: -R_B \frac{4L}{5} + \frac{9L}{5} P = 0$

$R_A = \frac{5}{4} P \quad R_B = \frac{9}{4} P$

③ Snitta



$T(x) = \frac{5}{4} P$



$T(x) = -P$

2015 forts

$$T(x) = \frac{5}{4}P \quad \text{Störst}$$

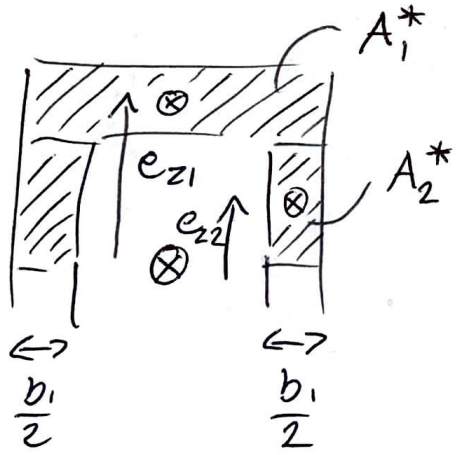
Bestäm böjskjuvspänning

$$\tau = \frac{T S_{A^*}}{I_y b}$$

Bestäm I_y

$$I_y = \frac{4a(6a)^3}{12} - \frac{2a(4a)^3}{12} = \frac{184a^4}{3}$$

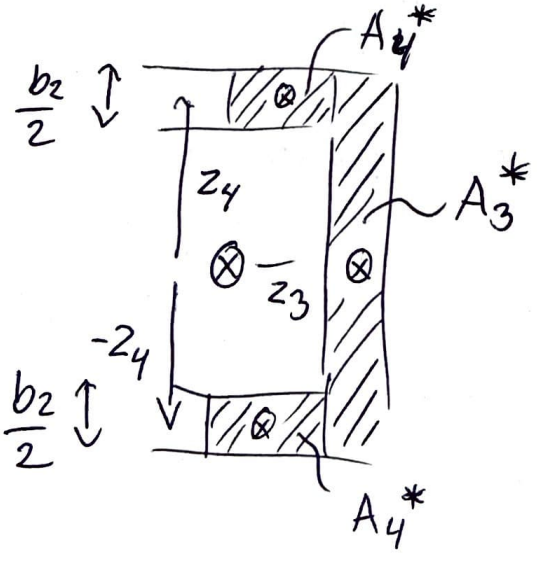
Bestäm S_{A^*}



$$S_{A^*_1} = e_{z1} A_1^* + e_{z2} A_2^*$$

$$= \frac{5}{2} a 4a^2 + 2a 2a^2 = 14a^3$$

$$b_1 = 2a$$



$$S_{A^*_2} = z_3 A_3^* + z_4 A_4^* - z_4 A_4^*$$

$$= 06a^2 = 0$$

$$b_2 = 2a$$

$$\Rightarrow \tau_{max} = \frac{\frac{5}{4}P 14a^3}{\frac{184}{3} a^4 2a} = \frac{105}{736}\frac{P}{a^2}$$